HISTOLOGY OF EYE LIDS CONJUNCTIVA & LACRIMAL APPARATUS

LECTURE OUTLINE

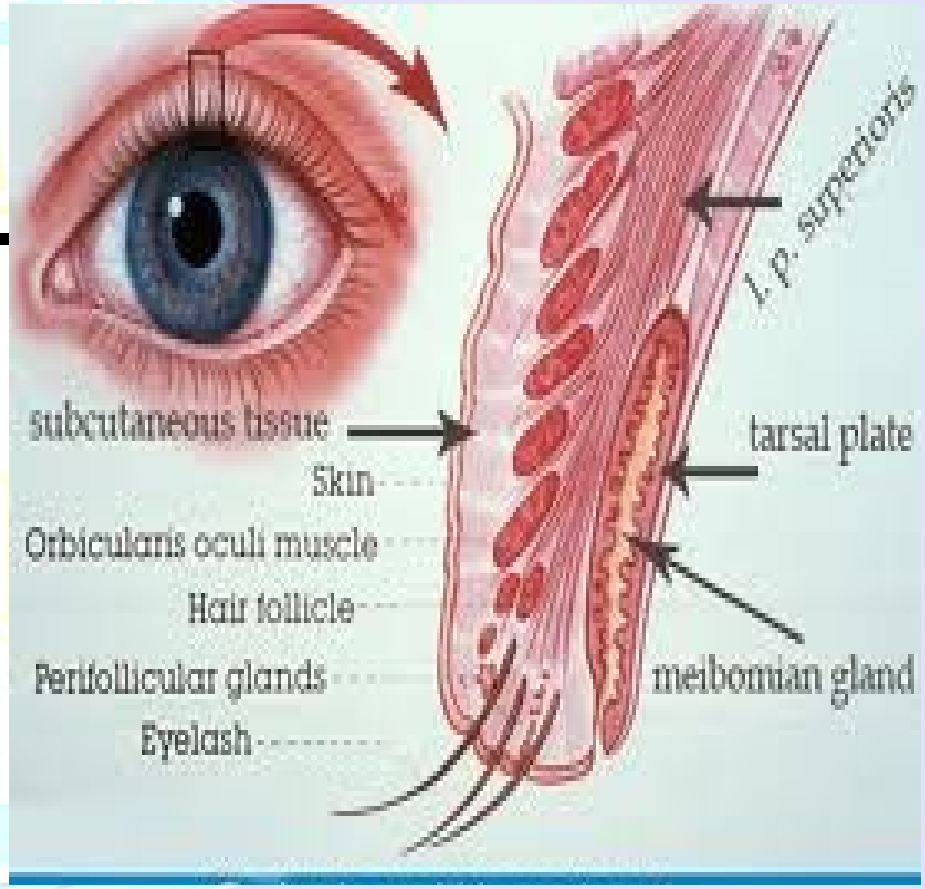
At the end of the lecture, student should be able to know the;

1) **Histological features of human eyelids.**
2) **Functions of different layers of eyelids.**
3) **Structural details of Conjunctiva.**
4) **Components of lacrimal apparatus.**
5) **Histological features of lacrimal gland.**
6) **Identify the slides of microscopic views of the above tissues.**

EYELIDS

They are movable folds, which

- **Protect the eyes from excessive light & injury.**
- **Also serve to lubricate the anterior portion of eyeballs.**
- **Upper eye lid is more movable than lower eye lid as it have levator palpebrae superioris muscle & Muller's muscle .**
- **The upper eyelid skin crease**



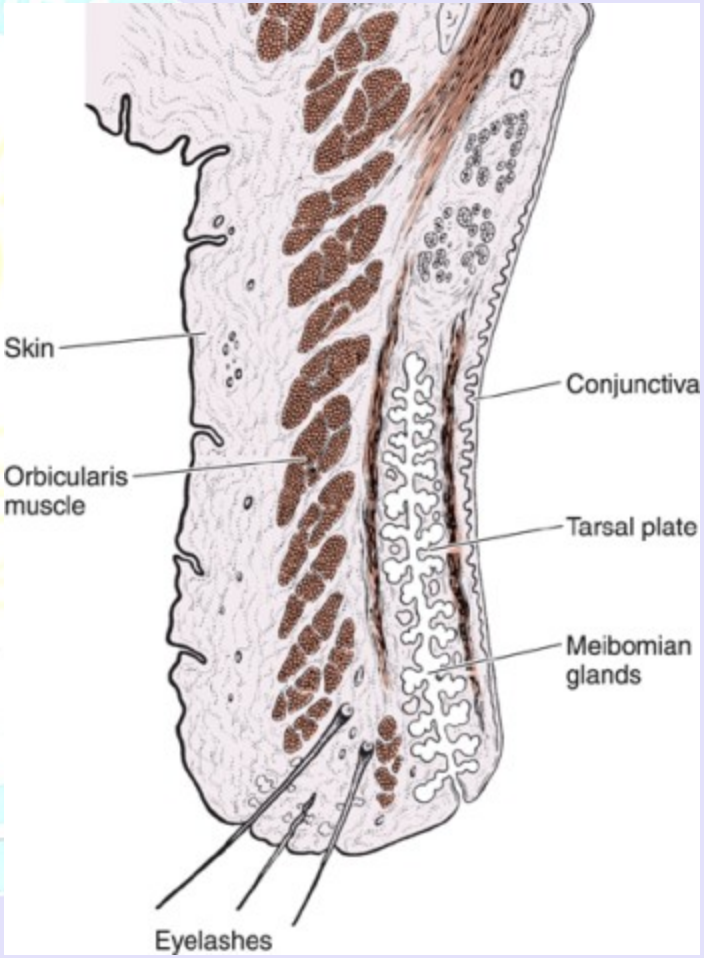
EYELIDS

- **Structurally: Eyelids consist of:**

1. **Skin: is extremely thin and contains small hairs & sweat glands.**
2. **Subcutaneous layer : loose CT, rich in elastic fibers but there is no fat.**
3. **Lid margin: there are 3 or 4 rows of long, stiff & curved hairs, which are called "eyelashes".**

   These hairs are associated with large sebaceous, glands called "glands of Zeis".

   Between the eyelashes, large, spiral sweat glands called "glands of Moll".

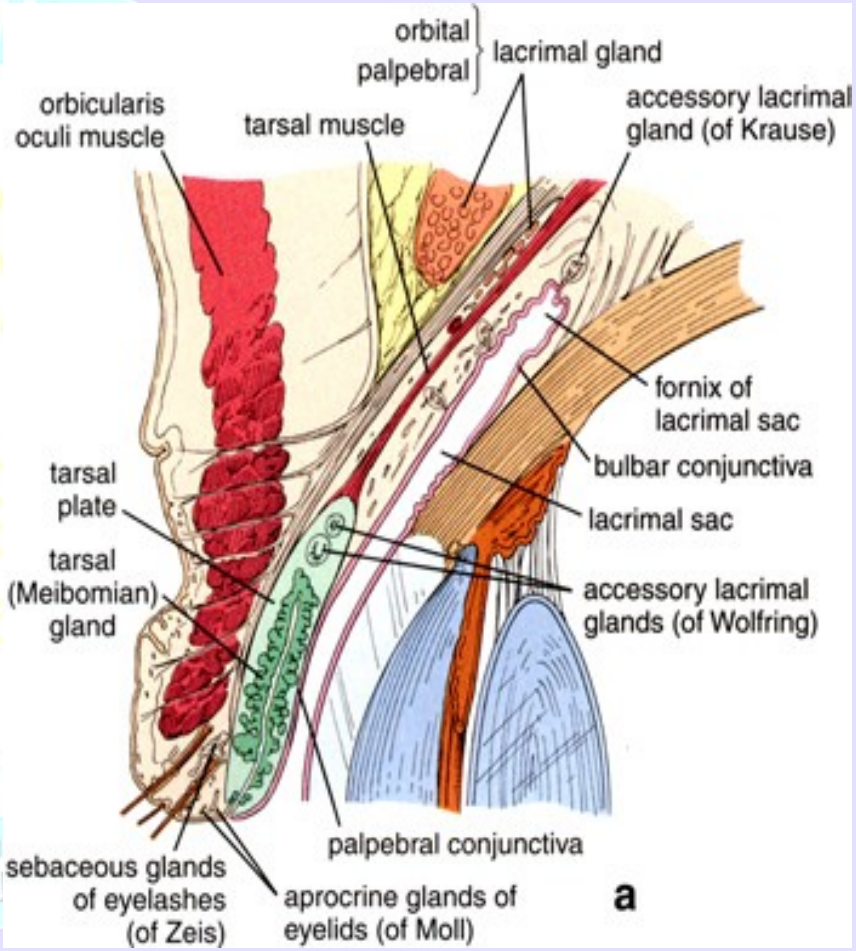


EYELIDS

4. A layer **of skeletal muscle** fibers is present beneath the skin. It is a part of orbicularis oculi.

5. **Tarsal plate**: lies next to the muscle layer.

- It is a curved plate of dense fibrous CT.
- a single row of very large sebaceous glands known as "tarsal or Meibomian glands." is embedded in the tarsal plate

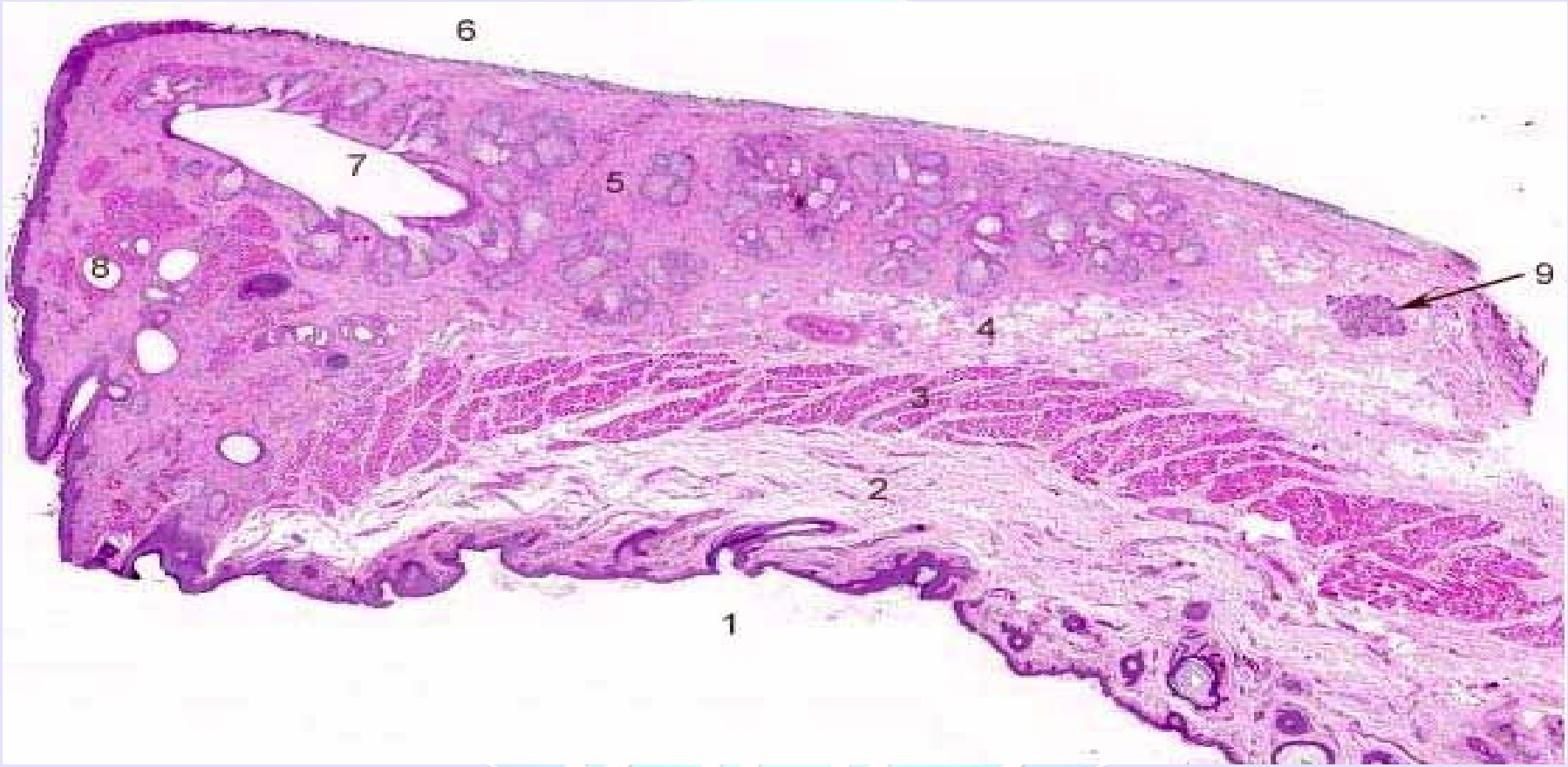


EYELIDS

- Ducts of Meibomian glands open at the margin
- Each tarsal gland consists of numerous alveoli, which open into a central straight duct.
- Secretion of these glands creates an oily layer on the surface of the tear film. This help to prevent rapid evaporation of the normal tear layer.

6. Conjunctiva : covers the posterior (inner) surface of the eyelid

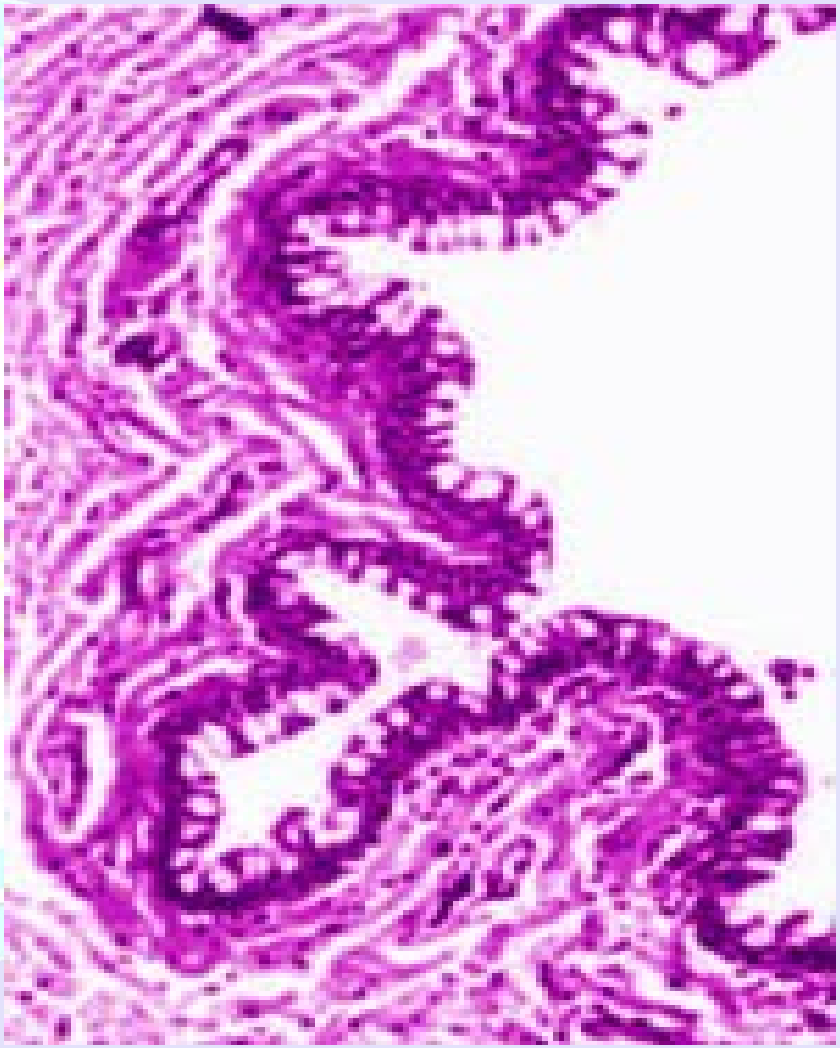
EYELID



- 1- skin surface; 2 fat and fascia; 3 orbicularis oculi; 4 levator aponeurosis; 5 tarsal plate; 6 conjunctival surface; 7 Meibomian ducts; 8 glands of Moll; 9 accessory lacrimal gland of Wolfring

CONJUNCTIVA

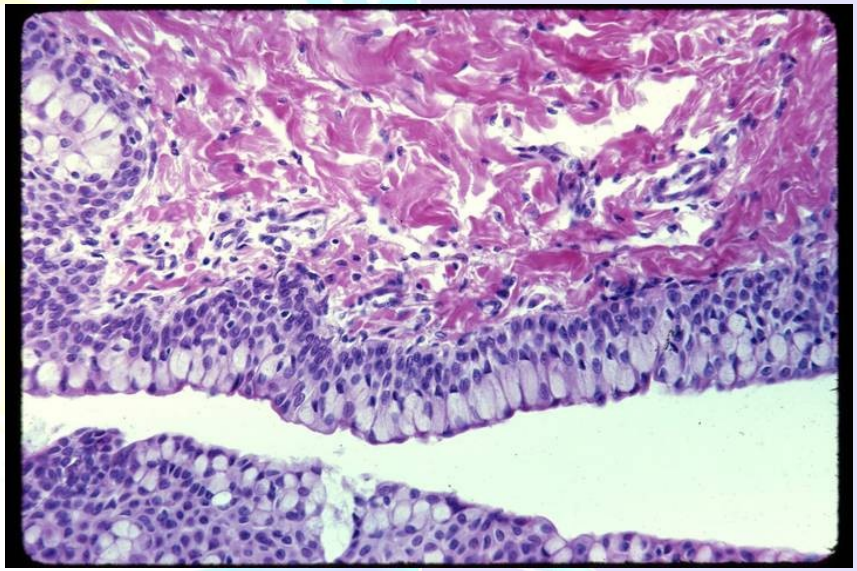
- It is a thin transparent mucous membrane that lines the eyelids from which it is reflected on to the anterior surface of the eyeball,
- covering it up to the corneal margin.



CONJUNCTIVA

- **Structure : consists of:**

1. **Epithelium :**
   stratified columnar
   – 2 layers:
   -surface cells are tall columnar.
   - deeper cells are low cuboidal.



2. **Lamina propria :**
   a thin layer of fine CT, which is heavily infiltrated with lymphocytes.

LACRIMAL APPARATUS

- The structures that secrete and drain tears from the eye
- It consists of:

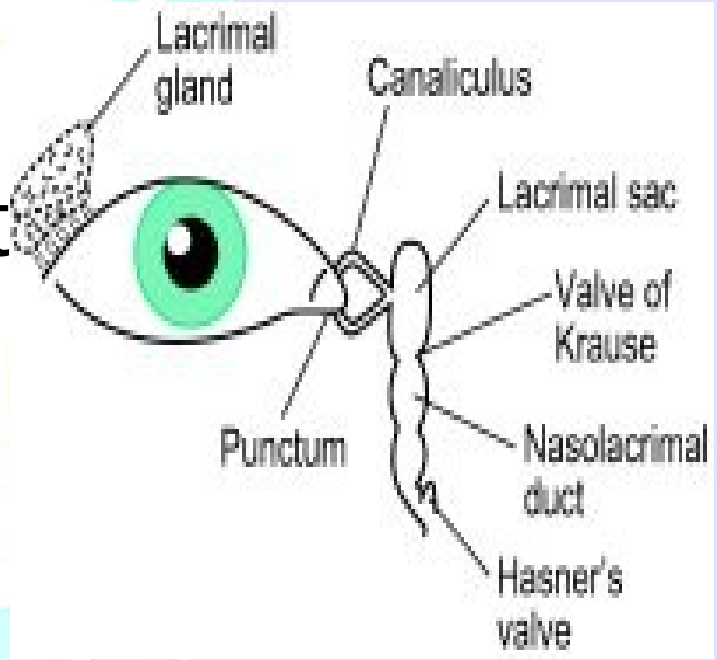
(a) The lacrimal gland, which secretes the t

(b) Lacrimal puncta

(c) Lacrimal canaliculli

(d) Lacrimal sac

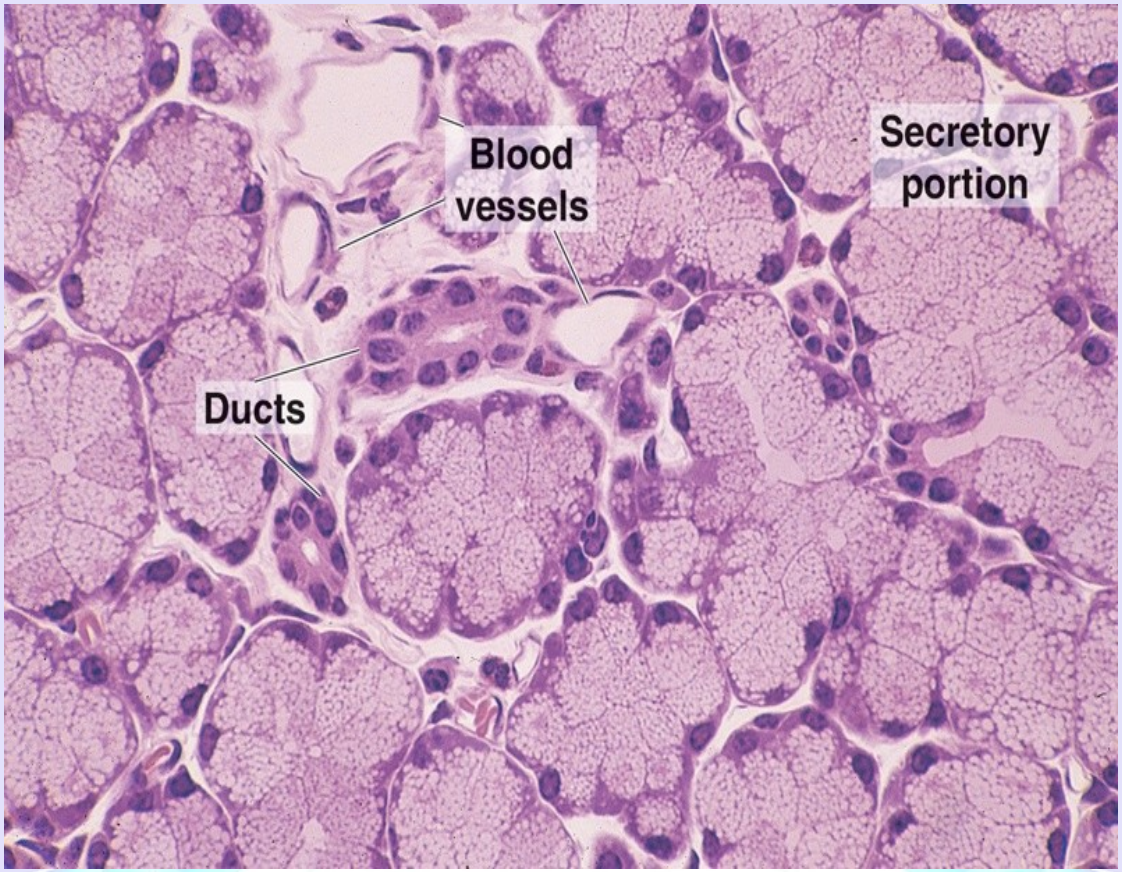
(e) Nasolacrimal duct



LACRIMAL GLAND

- They are compound tubuloalveolar glands.
- Purely serous type.
- Each gland consists of several separate lobes, which drain through 10 to 15 excretory ducts into the superior conjunctival fornix.

LACRIMAL GLAND



Photomicrograph of a section of a lacrimal gland.

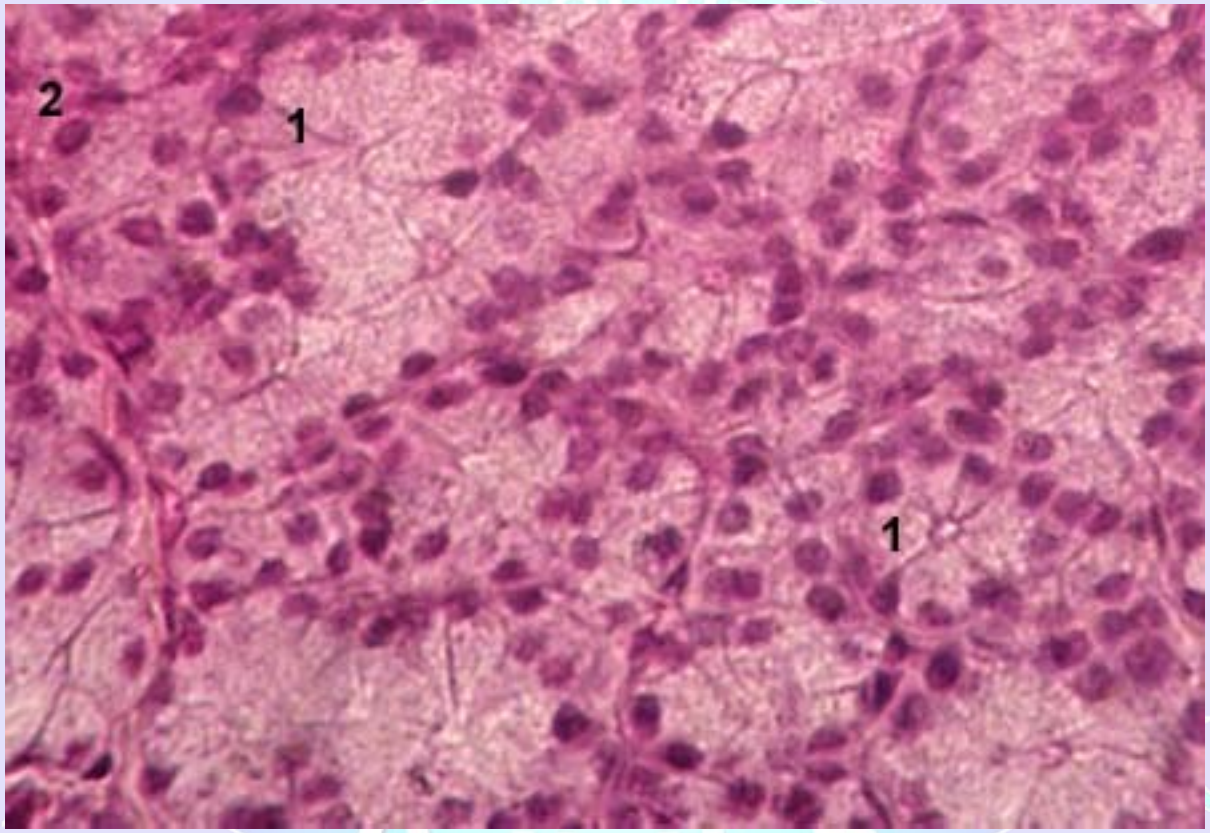
Note the sections of the tubuloalveolar secretory portions, excretory ducts, and blood vessels.

The secretory cells contain only small amounts of RNA (basophilia). They produce secretory material poor in proteins. H&E stain.

LACRIMAL GLAND

- Alveoli of the gland are lined by pyramidal cells, which show lightly stained apical secretary granules.
- Myoepithelial cells occur between the alveolar cells and their basal lamina.
- Stroma of the gland consists of loose CT.
- Ducts of the gland are lined by simple cuboidal epithelium.

LACRIMAL GLAND



Stained with haematoxylin and eosin

1 - secretory (end) unit 2 - intralobular excretory duct

SELF ASSESSMENT

Q1.Name the layers of eyelid.

Q2.Identify the epitelium shown in diagram, to which structure it belongs.



SELF ASSESSMENT

Q3.Name the components of lacrimal apparatus.

Q4.Which type of epithelium present in lacrimal gland?

THANK YOU

